



**Academic Year:**2026-2027

**Sem:** V

**Course:** Data Visualization Lab

**Course code:** TE7663

## **Practical No. 5**

**Title:** To design and implement a mini capstone project demonstrating data storytelling and dashboard theming techniques

**Software/ Tool Used:** Microsoft Power BI Desktop, Windows 10/11

**Data Set Used:** Any dataset

### **Objective**

- Identify a suitable real-world dataset and formulate meaningful analytical questions.
- Perform basic data cleaning, transformation, and preparation for visualization.
- Select appropriate charts and visual representations based on the nature of the data.
- Apply data storytelling principles to communicate insights effectively.
- Develop an interactive dashboard containing multiple coordinated visualizations.
- Apply dashboard theming, formatting, layout, and visual hierarchy techniques.
- Interpret the results and communicate actionable insights derived from the dashboard.

**Outcomes:**After successful completion of this experiment, students will be able to

- Prepare and organize a real-world dataset for visualization.
- Select and implement suitable visualizations for different analytical requirements.
- Develop a coherent data story using visual and textual elements.
- Apply dashboard layout, formatting, filtering, and theming techniques.
- Present insights and recommendations supported by visual evidence.

**Aim:**To develop an interactive data visualization dashboard that communicates meaningful insights from a real-world dataset using appropriate visualizations, data storytelling principles, interactive features, and a consistent dashboard theme..

### **Theory:      Data Storytelling**

Data storytelling is the process of communicating insights from data through a combination of data, visualizations, and narrative. A good data story does not merely display charts; it explains what the data means, why the observed pattern is important, and what action or conclusion can be derived from it.



A typical data story consists of three major elements:

Data + Visualization + Narrative → Insight

- Data provides the evidence.
- Visualization makes patterns, trends, relationships, and comparisons easier to understand.
- Narrative provides context and guides the audience towards the key message.

For example, instead of displaying monthly sales figures in a table, a line chart can show the sales trend over time. A narrative can then highlight the period of maximum growth, identify possible reasons, and communicate the business implication.

### **Data Storytelling Process**

A data storytelling process generally follows these stages:

1. Understand the problem → 2. Explore the data → 3. Identify insights → 4. Select visualizations → 5. Develop narrative → 6. Design dashboard → 7. Communicate findings

Students should first understand the problem statement and identify the target audience. The dataset should then be explored to identify trends, patterns, outliers, relationships, and significant differences.

The important insights should be converted into appropriate visualizations. These visualizations should subsequently be arranged into a logical sequence that guides the viewer through the story.

### **Choosing Appropriate Visualizations**

The choice of visualization should depend on the analytical purpose.

| <b>Purpose</b>        | <b>Suitable Visualization</b> |
|-----------------------|-------------------------------|
| Comparison            | Bar chart / Column chart      |
| Trend over time       | Line chart / Area chart       |
| Distribution          | Histogram / Box plot          |
| Relationship          | Scatter plot                  |
| Part-to-whole         | Pie / Donut / Stacked bar     |
| Geographical analysis | Map                           |
| Ranking               | Sorted bar chart              |
| KPI monitoring        | KPI card / Gauge              |
| Multiple dimensions   | Heatmap / Treemap             |



Students should avoid selecting charts merely because they appear visually attractive. The visualization must support the intended analytical message.

## **Dashboard**

A dashboard is a visual interface that presents important information, metrics, and insights in a consolidated and interactive format.

An effective dashboard should:

- Present important KPIs prominently.
- Provide a logical arrangement of visualizations.
- Minimize unnecessary visual elements.
- Allow users to filter or interact with the data where appropriate.
- Maintain consistency in fonts, colours, labels, and chart styles.
- Enable users to understand the major insights quickly.

A dashboard should answer questions such as:

- What is happening?
- Where is it happening?
- Why might it be happening?
- What are the important trends or exceptions?
- What action can be considered?

## **Dashboard Theming**

Dashboard theming refers to the systematic application of visual design elements across a dashboard to create a consistent and professional appearance.

Important elements of dashboard theming include:

- Colour Scheme
- Colours should be selected purposefully. A limited and consistent colour palette should be used throughout the dashboard.

For example:

- One colour can represent the primary metric.
- Another can highlight positive or negative changes.
- Neutral colours can be used for supporting information.



Colours should not be used excessively because too many colours can make the dashboard difficult to interpret.

### **Typography**

- Fonts should be consistent across the dashboard. Different font sizes can be used to establish hierarchy:
- Dashboard Title → Section Heading → KPI → Chart Title → Supporting Text

### **Layout**

Visual elements should be arranged according to their importance. The most important KPIs and insights should receive greater visual prominence.

A typical dashboard layout can be:

Title / Context



KPI Cards



Primary Trend / Key Visualization



Supporting Visualizations



Filters / Detailed Analysis



Key Insights / Conclusion

### **Visual Hierarchy**

Visual hierarchy determines the order in which users perceive and interpret information.

Students should use:

- Size
- Position
- Colour
- Contrast
- Whitespace
- Typography

to establish hierarchy.

Important information should generally be positioned where the user is likely to notice it first. Unnecessary borders, decorations, excessive icons, and distracting backgrounds should be avoided.

- **Interactivity**

Interactive features allow users to explore different aspects of the data.



Examples include:

- Filters
- Slicers
- Dropdown selections
- Drill-down
- Tooltips
- Cross-filtering
- Interactive legends
- Date-range selectors

Interactivity should serve a clear analytical purpose. Students should avoid adding interactive components simply for decoration.

- **Narrative Structure of a Dashboard**

Students should structure their dashboard as a story rather than as a collection of unrelated charts.

A suitable narrative structure is:

Context → Overview → Exploration → Key Insight → Conclusion/Action

For example:

Context: What problem is being analysed?

Overview: What does the overall data indicate?

Exploration: What trends, differences, or relationships exist?

Key Insight: What is the most important finding?

Action: What conclusion or recommendation can be drawn?

### **Project Guidelines**

Students shall develop a mini capstone project using a real-world dataset. The project should demonstrate both data storytelling and dashboard theming.

Step 1: Select a Domain

Students may select a domain such as:

- Business and Sales
- Healthcare
- Education
- Banking and Finance
- E-commerce
- Transportation
- Sports Analytics
- Agriculture
- Environment
- Social Media Analytics
- Human Resources
- Tourism
- Energy

The selected dataset should contain sufficient attributes to support meaningful analysis.



## **Data Preparation**

Students should:

1. Load the dataset.
2. Examine the attributes and data types.
3. Identify missing values.
4. Remove or handle duplicate records.
5. Identify outliers where appropriate.
6. Perform necessary data transformations.
7. Create calculated fields/measures if required.

Students should document the major preprocessing steps.

## **Exploratory Data Analysis**

Perform exploratory analysis to identify:

- Trends
- Patterns
- Relationships
- Comparisons
- Distributions
- Outliers
- Significant categories
- Changes over time

The objective is to identify the **story hidden within the data**.

## **Develop the Dashboard**

The dashboard should contain, wherever relevant:

- **Dashboard title**
- **Project context/problem statement**
- **At least 3 KPI indicators**
- **At least 4 meaningful visualizations**
- Appropriate filters/slicers
- Descriptive chart titles
- Legends and labels
- Source information
- Key findings/insights

Students should select visualizations based on the analytical questions rather than using a fixed set of charts.

## **Dashboard Theming Guidelines**

Students must demonstrate deliberate use of dashboard theming.

The dashboard should maintain:

### **Consistency**

Use a consistent:

- Colour palette
- Font family
- Font sizes
- Chart style
- Border style
- Alignment



## **Contrast**

Ensure sufficient contrast between:

- Background and text
- Chart elements and background
- Important and supporting information

## **Simplicity**

Avoid:

- Excessive colours
- 3D charts
- Unnecessary decorative graphics
- Excessive gridlines
- Cluttered layouts
- Unnecessary animations

## **Branding/Theme**

Students may create a theme based on the selected domain.

For example:

- Healthcare → clean and professional visual theme
- Finance → formal analytical theme
- Agriculture → nature-oriented theme
- Sports → energetic analytical theme

The theme should **support the story rather than distract from the data**.

## **Data Storytelling Requirements**

The final dashboard should communicate a clear story.

Students should identify:

### **What?**

What does the data represent?

### **So What?**

What important patterns or insights have been identified?

### **Now What?**

What conclusion, recommendation, or decision can be derived from the analysis?

The final dashboard should allow a user unfamiliar with the dataset to understand the **main message within a short period of time**.



## **Expected Project Deliverables**

Students should submit:

1. **Dataset**
2. **Problem statement**
3. **Project objectives**
4. **Data preprocessing documentation**
5. **Exploratory analysis**
6. **Final interactive dashboard**
7. **Dashboard theme/design description**
8. **Key insights**
9. **Conclusions/recommendations**
10. **Screenshots of the final dashboard**
11. **Project report**
12. **Presentation/demo**



**Academic Year:**2026-2027

**Sem:** V

**Course:** Data Visualization Lab

**Course code:** TE7663

**Name:**

**Roll No:**

**Assignment questions:**

1. Explain the concept of data storytelling. Describe the role of data, visualization, and narrative in communicating insights effectively.
2. Differentiate between data visualization and data storytelling. Explain how storytelling adds value to a dashboard.
3. Explain the importance of selecting an appropriate visualization for a given analytical question. Give suitable examples for comparison, trend, distribution, relationship, and part-to-whole analysis.
4. What is a dashboard? Explain the essential characteristics of an effective and user-friendly dashboard.
5. What is visual hierarchy? Explain how size, position, colour, contrast, and whitespace can be used to establish visual hierarchy in a dashboard.

**Practice questions:**

1. What is the primary purpose of data storytelling?  
A. To make dashboards visually complex  
B. To communicate meaningful insights from data effectively  
C. To display as many charts as possible  
D. To replace data analysis
2. Which combination best represents the three major components of data storytelling?  
A. Data + Code + Database  
B. Data + Visualization + Narrative  
C. Charts + Images + Animation  
D. Dataset + Programming + Statistics
3. Which visualization is generally most appropriate for showing a trend over time?  
A. Pie chart  
B. Line chart  
C. Treemap  
D. Donut chart
4. Which of the following is an important objective of dashboard theming?  
A. Increasing the number of colours  
B. Adding decorative elements to every chart  
C. Maintaining visual consistency across the dashboard  
D. Making every visualization different



5. Which visualization is most appropriate for analysing the relationship between two numerical variables?
- A. Scatter plot
  - B. Pie chart
  - C. Bar chart
  - D. Donut chart
6. Which of the following is an example of an interactive dashboard feature?
- A. Static image
  - B. Slicer/filter
  - C. Printed table
  - D. Dashboard background
7. What is the purpose of visual hierarchy in dashboard design?
- A. To make all elements appear equally important
  - B. To guide the user's attention towards important information
  - C. To increase the number of charts
  - D. To hide less important data
8. In the “What? – So What? – Now What?” approach, “So What?” primarily focuses on:
- A. Dataset collection
  - B. Data cleaning
  - C. Interpretation and significance of the findings
  - D. Selection of dashboard software
9. Which of the following is NOT a recommended dashboard design practice?
- A. Using consistent fonts
  - B. Maintaining sufficient contrast
  - C. Using excessive colours and decorative elements
  - D. Providing meaningful chart titles
10. What should be the primary consideration when selecting a visualization?
- A. The most colourful chart available
  - B. The chart that occupies the most space
  - C. The analytical question and type of data
  - D. The number of charts already present on the dashboard